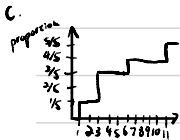


Homework: Categorical and Numeric Variables

Question 1:

		Red	Blue	Green	Grey
a. good	1	1	0	0	0
	2	0	1	0	0
	3	0	1	0	0
	4	1	0	0	0
	5	0	0	1	0
	6	0	0	0	1

Color	$\hat{p}$
b. Red	2/6
Blue	2/6
Green	1/6
Grey	1/6



Problem 2:

a. $F_Y(y) = P(Y \leq y) = P(a + bX \leq y) = P(X \leq \frac{y-a}{b}) = F_X(\frac{y-a}{b})$ *Use $F_X(x)$ but transform y to $\frac{y-a}{b}$

b. $\frac{1+3+4}{3} = \frac{8}{3} = \bar{X} = \frac{8}{3}$

c. $\frac{1^2+3^2+4^2}{3} = \frac{1+9+16}{3} = \frac{26}{3} = \overline{X^2} = \frac{26}{3}$

d. $f_Y(y) = \left[F_X(\frac{y-a}{b}) \right]' = F_X'(\frac{y-a}{b}) \cdot \left(\frac{y-a}{b} \right)' dy$ change $f_X(\frac{y-a}{b})$

$f_Y(y) = f_X(\frac{y-a}{b}) \cdot \left(\frac{1}{b} \right)$ $\frac{1}{b}y - \frac{1}{b}a \xrightarrow{dy} \frac{1}{b}$

d. $\left(\frac{8}{3} \right)^2 = \frac{64}{9} \approx 7.1$ This does not equal the mean of the square from part c.

Problem 3

a. Density



$$f_X(x) = F_X(x)$$

$$= (1 - e^{-x})'$$

$$= -(-e^{-x})$$

$$e^{-x} = \frac{1}{e^x}$$

$$f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Distribution



C (continued).

b. $P(X \geq 3) =$

Distribution

$$1 - P(X \leq 3)$$

$$P(X \leq 3) = F_X(3) - F_X(0)$$

$$= 1 - e^{-3} - 0$$

$$= 1 - e^{-3}$$

$$1 - (1 - e^{-3})$$

$$= \boxed{e^{-3}}$$

$$P(X \leq 35)$$

$$F(35) - F(0)$$

$$\boxed{1 - e^{-35}}$$

$$F_X\left(\frac{y-a}{b}\right)$$

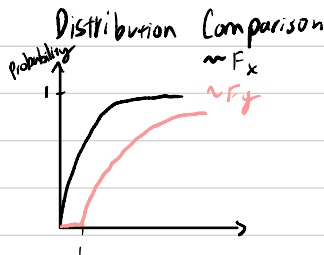
$$y = 1 + 3X$$

$$c. F_Y(y) = F_X\left(\frac{y-a}{b}\right) = F_X\left(\frac{y-1}{3}\right)$$

$$= \boxed{1 - e^{-\left(\frac{y-1}{3}\right)}}$$

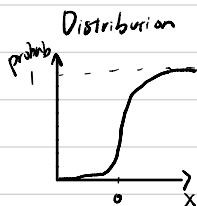
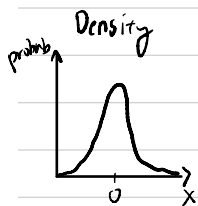
$$P[Y \leq 10] = 1 - e^{-\left(\frac{10-1}{3}\right)}$$

$$= \boxed{1 - e^{-3}}$$



Problem 4

a.



$$\frac{1}{1+e^0} = \frac{1}{2}$$

$$\frac{1}{1+e^{-1/2}}$$

$$\frac{1}{1+e^2}$$

$$f(x) = \frac{1}{1+e^{-x}}$$

$$F(x) = (1+e^{-x})^{-1}$$

$$F'(x) = -(1+e^{-x})^{-2} \cdot (-e^{-x})$$

$$\frac{1}{1+e} \cdot \frac{1}{1+e^2}$$

$$= (1+e^{-x})^{-2} (e^{-x})$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

b. $P[X \geq .8] =$

$$1 - P[X \leq .8]$$

$$1 - \frac{1}{1+e^{0.8}}$$

$P[X \leq .3] =$

$$\frac{1}{1+e^{-0.3}}$$

c. $Y = 2X - 1$

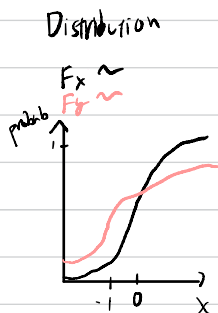
$$F_X(x) = \frac{1}{1+e^{-x}}$$

$$F_Y(y) = F_X\left(\frac{y-a}{b}\right) = F_X\left(\frac{y+1}{2}\right) =$$

$$F_Y(y) = \frac{1}{1+e^{-\frac{(y+1)}{2}}}$$

$$P[Y \leq 0] = \frac{1}{1+e^{-\frac{(0+1)}{2}}}$$

$$\frac{1}{1+e^{-\frac{1}{2}}}$$



Problem 5

a. median = $F_x(x) = 0.5$

d.

$$P[X \leq x] = 0.5$$

$$\frac{1}{1+e^{-x}} = u$$

$$1 - e^{-x} = 0.5$$

$$0.5 = e^{-x}$$

$$1 = u(1+e^{-x})$$

$$\ln(0.5) = -x \ln(e)$$

$$\frac{1}{u} = 1+e^{-x}$$

$$-\ln(0.5) = x$$

$$-1 \cdot \ln(0.5) \rightarrow \ln(0.5) \rightarrow$$

$$\boxed{x = \ln(2)} \quad \frac{1}{u} - 1 = e^{-x}$$

$$\ln\left(\frac{1}{u} - 1\right) = -x \ln(e)$$

b. $\frac{1}{1+e^x} = 0.5$

$$\boxed{\ln\left(\frac{1}{u} - 1\right) = x}$$

$$1 = 0.5(1+e^{-x})$$

$$2 = 1+e^{-x}$$

$$1 = e^{-x}$$

$$\ln(1) = -x \ln(e)$$

$$0 = -x$$

$$\boxed{0 = x}$$

c. $F(x) = u$

$$1 - e^{-x} = u$$

$$1 - u = e^{-x}$$

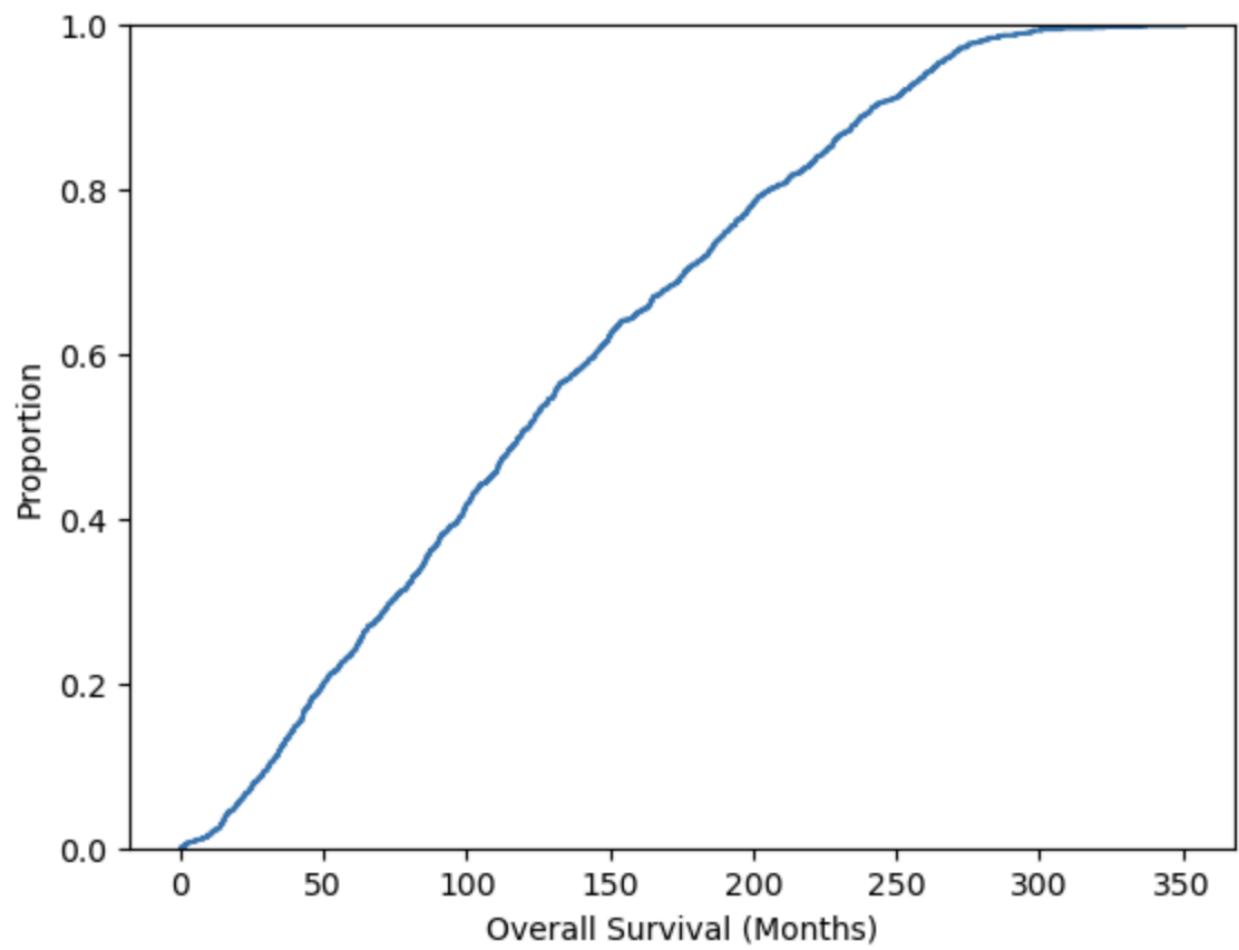
$$\ln(1-u) = -x \ln(e)$$

$$\ln(1-u) = -x$$

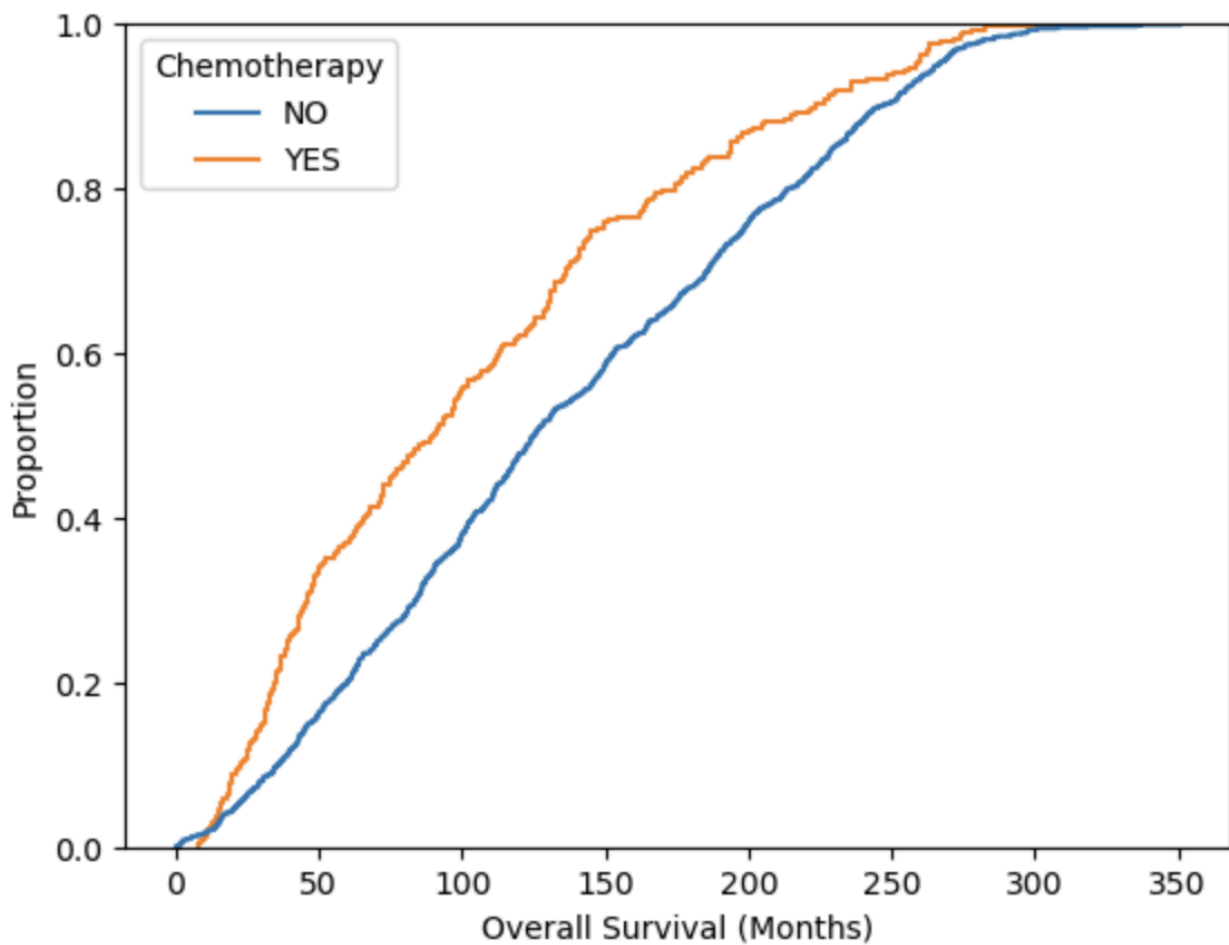
$$-\ln(1-u) = x \rightarrow F^{-1}(u) = x$$

$$\boxed{\ln(1-u^{-1}) = x \rightarrow F'(u) = x}$$

a.



b.



We predict that conditional on receiving chemotherapy, the patients will live a shorter time. The YES line is more bowed out compared to the NO line indicating how a larger proportion of the chemotherapy group survived less months compared to the no chemotherapy group. For example, at 0.4, the chemotherapy group is at around 50 survival months while no chemotherapy survived for 100 months. As the yes chemotherapy line is bowed out, the slope is steeper and thus accumulating more of the proportion in the lower months.

c. Chemotherapy does not seem to be an effective treatment looking at the EDCF. However, you cannot conclude based off of this graph alone that chemotherapy is not an effective treatment as other survival factors are not accounted for. However, it seems that chemotherapy is not associated with higher months of survival as seen in the EDCF as a greater proportion survives less months compared to the no chemotherapy group.